

Data Cleaning

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```
library(here)
```

```
## here() starts at /Users/flannerymcnair/Documents/710_Stats/ENV710_Group12_Final_Project
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.0      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr   1.6.0
```

```
## v ggplot2    4.0.2      v tibble    3.3.1
```

```
## v lubridate  1.9.5      v tidyr     1.3.2
```

```
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

```
library(lubridate)
```

```
library(ggpubr)
```

```
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

Uploading Data

```
SVI_2022 <- read_csv(here("California_county_SVI_2022.csv"))
```

```
## Rows: 58 Columns: 158
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr   (7): ST, STATE, ST_ABBR, STCNTY, COUNTY, FIPS, LOCATION
```

```
## dbl (151): AREA_SQMI, E_TOTPOP, M_TOTPOP, E_HU, M_HU, E_HH, M_HH, E_POV150, ...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
SVI_2020 <- read_csv(here("California_county_SVI_2020.csv"))
```

```
## Rows: 58 Columns: 158
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr   (7): ST, STATE, ST_ABBR, STCNTY, COUNTY, FIPS, LOCATION
```

```
## dbl (151): AREA_SQMI, E_TOTPOP, M_TOTPOP, E_HU, M_HU, E_HH, M_HH, E_POV150, ...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
SVI_2018 <- read_csv(here("California_county_SVI_2018.csv"))
```

```
## Rows: 58 Columns: 123
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (6): ST, STATE, ST_ABBR, COUNTY, FIPS, LOCATION
```

```
## dbl (117): AREA_SQMI, E_TOTPOP, M_TOTPOP, E_HU, M_HU, E_HH, M_HH, E_POV, M_P...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
SVI_2016 <- read_csv(here("California_county_SVI_2016.csv"))
```

```
## Rows: 58 Columns: 123
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (6): ST, STATE, ST_ABBR, COUNTY, FIPS, LOCATION
```

```
## dbl (117): AREA_SQMI, E_TOTPOP, M_TOTPOP, E_HU, M_HU, E_HH, M_HH, E_POV, M_P...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
asthma <- read_csv(here("current-asthma-prevalence-by-county-2015_2022.csv"))
```

```
## Rows: 1652 Columns: 8
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (7): COUNTY, YEARS, STRATA, AGE GROUP, 95% CONFIDENCE INTERVAL, COUNTIES...
```

```
## dbl (1): CURRENT PREVALENCE
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
AQI_2022 <- read_csv(here("daily_aqi_by_county_2022.csv"))
```

```
## Rows: 324419 Columns: 10
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (7): State Name, county Name, State Code, County Code, Category, Defini...
```

```
## dbl (2): AQI, Number of Sites Reporting
```

```
## date (1): Date
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
AQI_2020 <- read_csv(here("daily_aqi_by_county_2020.csv"))
```

```
## Rows: 325138 Columns: 10
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (7): State Name, county Name, State Code, County Code, Category, Defini...
```

```
## dbl (2): AQI, Number of Sites Reporting
```

```
## date (1): Date
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
AQI_2018 <- read_csv(here("daily_aqi_by_county_2018.csv"))

## Rows: 327541 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (7): State Name, county Name, State Code, County Code, Category, Defini...
## dbl (2): AQI, Number of Sites Reporting
## date (1): Date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
AQI_2016 <- read_csv(here("daily_aqi_by_county_2016.csv"))
```

```
## Rows: 321071 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (7): State Name, county Name, State Code, County Code, Category, Defini...
## dbl (2): AQI, Number of Sites Reporting
## date (1): Date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Cleaning SVI Data

```
SVI_2022_clean <- SVI_2022 %>%
  subset(select = -c(STATE, ST_ABBR, ST, LOCATION)) %>%
  select(-starts_with("M_")) %>%
  select(-starts_with("MP_")) %>%
  select(-starts_with("F_"))
SVI_2022_clean$COUNTY <- gsub(" County", "", SVI_2022_clean$COUNTY)

SVI_2020_clean <- SVI_2020 %>%
  subset(select = -c(STATE, ST_ABBR, ST, LOCATION)) %>%
  select(-starts_with("M_")) %>%
  select(-starts_with("MP_")) %>%
  select(-starts_with("F_"))
SVI_2020_clean$COUNTY <- gsub(" County", "", SVI_2020_clean$COUNTY)

SVI_2018_clean <- SVI_2018 %>%
  subset(select = -c(STATE, ST_ABBR, ST, LOCATION)) %>%
  select(-starts_with("M_")) %>%
  select(-starts_with("MP_")) %>%
  select(-starts_with("F_"))
SVI_2018_clean$COUNTY <- gsub(" County", "", SVI_2018_clean$COUNTY)

SVI_2016_clean <- SVI_2016 %>%
  subset(select = -c(STATE, ST_ABBR, ST, LOCATION)) %>%
  select(-starts_with("M_")) %>%
  select(-starts_with("MP_")) %>%
  select(-starts_with("F_"))
SVI_2016_clean$COUNTY <- gsub(" County", "", SVI_2016_clean$COUNTY)
```

Cleaning Asthma data

```

asthma_clean <- asthma %>%
  filter(`AGE GROUP` == "All ages")

years <- sapply(strsplit(asthma_clean$YEARS, "-"), `[`, 2)

asthma_clean <- asthma_clean %>%
  mutate(YEARS = years) %>%
  select(c(COUNTY, YEARS, `CURRENT PREVALENCE`, `95% CONFIDENCE INTERVAL`))

AQI_2022$Date <- as.Date((AQI_2022$Date))
AQI_2022_clean <- AQI_2022 %>%
  filter(`State Name` == "California") %>%
  aggregate(AQI ~ `county Name`, FUN = "mean") %>%
  rename(COUNTY = `county Name`)

AQI_2020$Date <- as.Date((AQI_2020$Date))
AQI_2020_clean <- AQI_2020 %>%
  filter(`State Name` == "California") %>%
  aggregate(AQI ~ `county Name`, FUN = "mean") %>%
  rename(COUNTY = `county Name`)

AQI_2018$Date <- as.Date((AQI_2018$Date))
AQI_2018_clean <- AQI_2018 %>%
  filter(`State Name` == "California") %>%
  aggregate(AQI ~ `county Name`, FUN = "mean") %>%
  rename(COUNTY = `county Name`)

AQI_2016$Date <- as.Date((AQI_2016$Date))
AQI_2016_clean <- AQI_2016 %>%
  filter(`State Name` == "California") %>%
  aggregate(AQI ~ `county Name`, FUN = "mean") %>%
  rename(COUNTY = `county Name`)

```

Combining SVI and Asthma data

```

asthma_2016 <- asthma_clean %>%
  filter(YEARS == "2016")

data_2016 <- inner_join(AQI_2016_clean, asthma_2016, by = "COUNTY")
data_2016_SVI <- inner_join(data_2016, SVI_2016_clean, by = "COUNTY")

asthma_2018 <- asthma_clean %>%
  filter(YEARS == "2018")

data_2018 <- inner_join(AQI_2018_clean, asthma_2018, by = "COUNTY")
data_2018_SVI <- inner_join(data_2018, SVI_2018_clean, by = "COUNTY")

asthma_2020 <- asthma_clean %>%
  filter(YEARS == "2020")

data_2020 <- inner_join(AQI_2020_clean, asthma_2020, by = "COUNTY")
data_2020_SVI <- inner_join(data_2020, SVI_2020_clean, by = "COUNTY")

asthma_2022 <- asthma_clean %>%

```

```
filter(YEARS == "2022")
```

```
data_2022 <- inner_join(AQI_2022_clean, asthma_2022, by = "COUNTY")
data_2022_SVI <- inner_join(data_2022, SVI_2022_clean, by = "COUNTY")
```

Plotting AQI vs asthma prevalence

```
plot_2016 <- ggplot(data_2016, aes(AQI, `CURRENT PREVALENCE`)) +
  geom_point() +
  geom_smooth(method = "lm") +
  stat_cor(aes(label = after_stat(rr.label))) +
  labs(y = "Asthma Prevalence (%)",
       title = "AQI vs Asthma Prevalence by California County in 2016")
```

```
plot_2016
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
```

```
## (`stat_smooth()`).
```

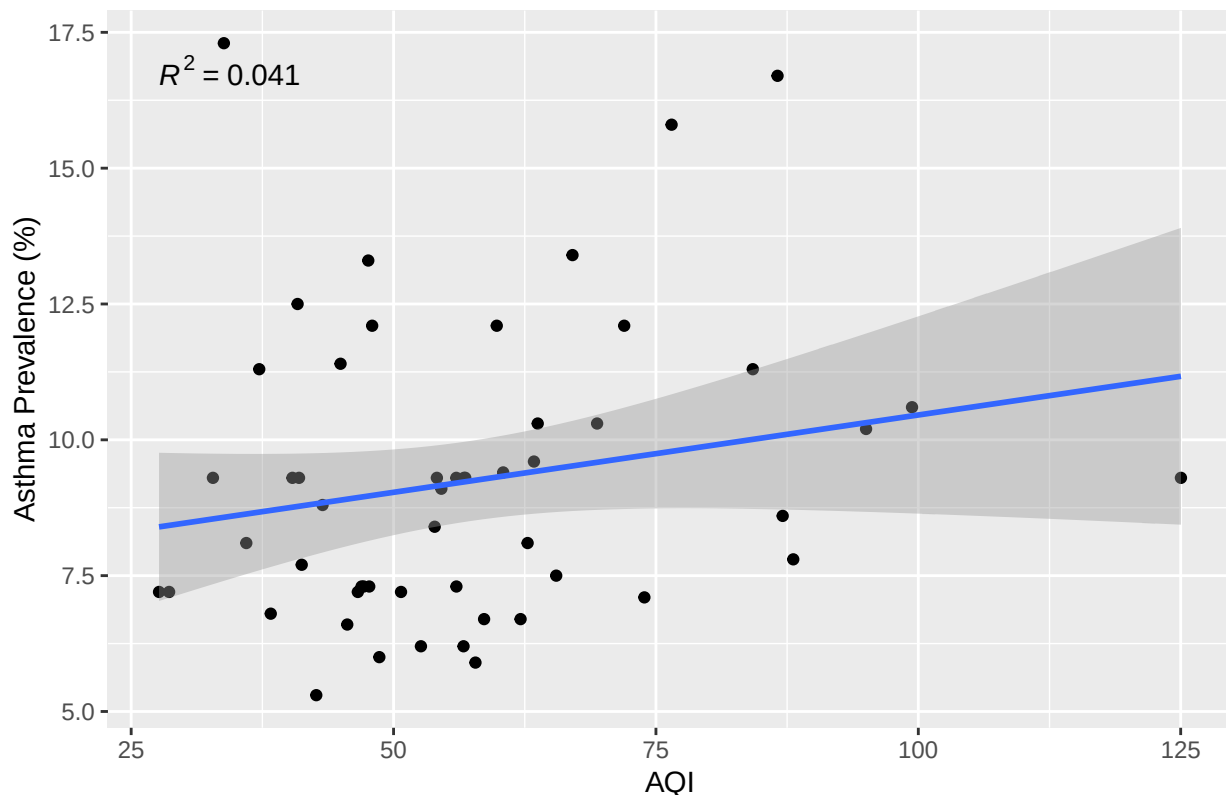
```
## Warning: Removed 1 row containing non-finite outside the scale range
```

```
## (`stat_cor()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```

AQI vs Asthma Prevalence by California County in 2016



```
mlm.2 <- lm(data_2016$AQI ~ data_2016$`CURRENT PREVALENCE`)
summary(mlm.2)
```

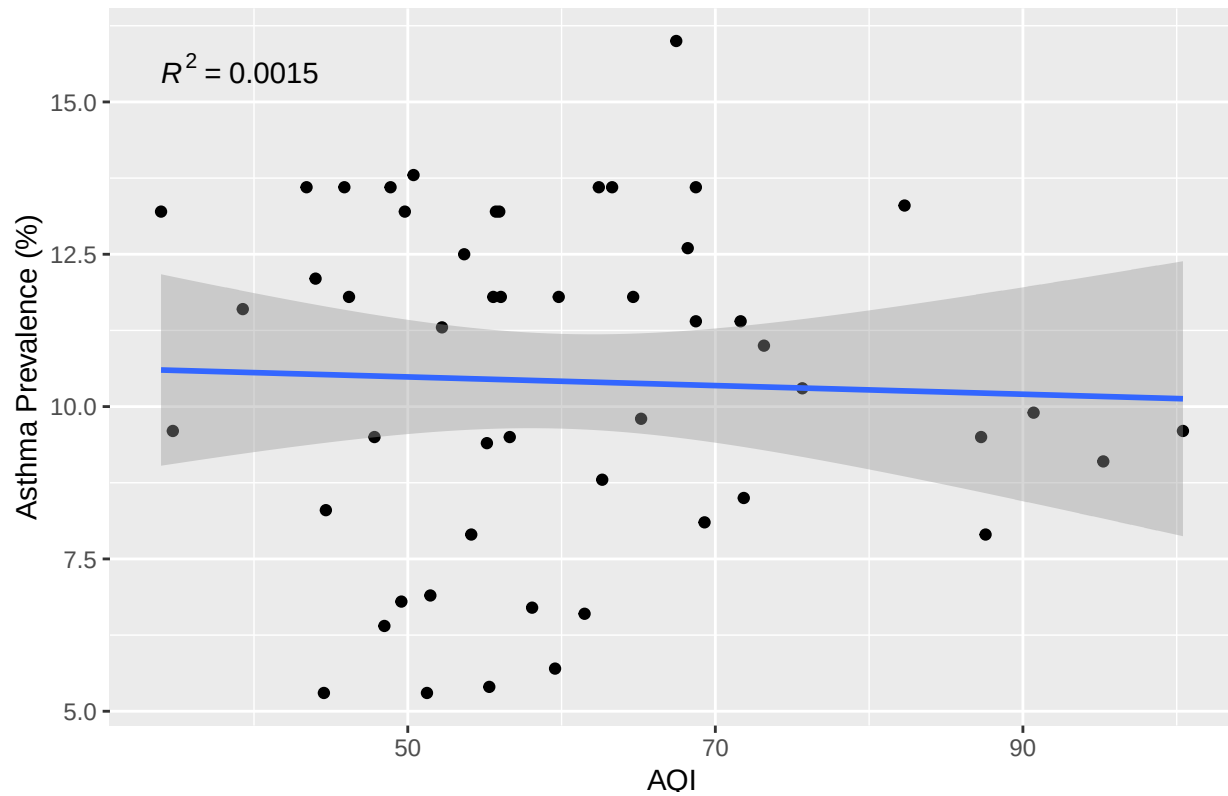
```
##
## Call:
## lm(formula = data_2016$AQI ~ data_2016$`CURRENT PREVALENCE`)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -34.956 -13.372  -1.501   7.174  67.705
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      43.9861     9.3627   4.698 2.02e-05 ***
## data_2016$`CURRENT PREVALENCE`  1.4337     0.9733   1.473   0.147
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.98 on 51 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.04081,    Adjusted R-squared:  0.02201
## F-statistic:  2.17 on 1 and 51 DF,  p-value: 0.1469

plot_2018 <- ggplot(data_2018, aes(AQI, `CURRENT PREVALENCE`)) +
  geom_point() +
  geom_smooth(method = "lm") +
  stat_cor(aes(label = after_stat(rr.label))) +
  labs(y = "Asthma Prevalence (%)",
       title = "AQI vs Asthma Prevalence by California County in 2018")

plot_2018

## `geom_smooth()` using formula = 'y ~ x'
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_smooth()`).
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_cor()`).
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

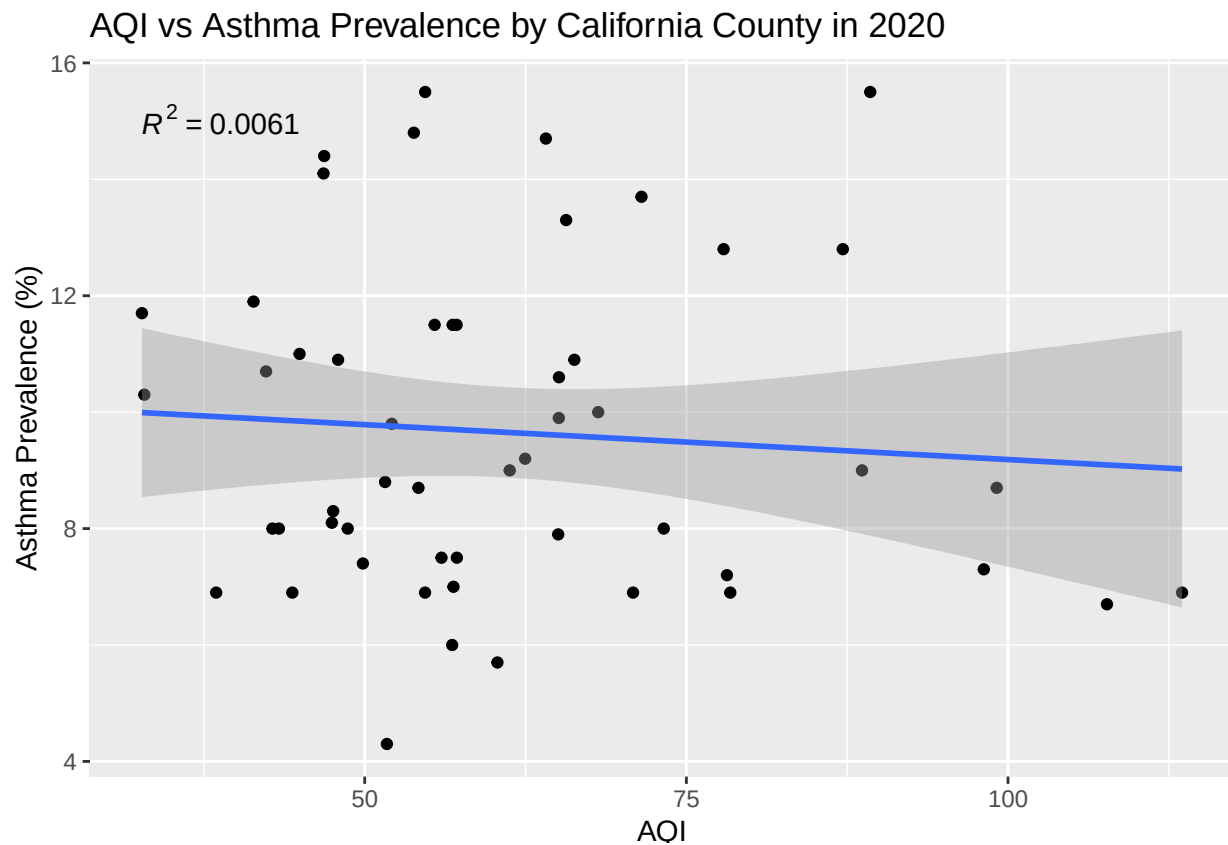
AQI vs Asthma Prevalence by California County in 2018



```
plot_2020 <- ggplot(data_2020, aes(AQI, `CURRENT PREVALENCE`)) +
  geom_point() +
  geom_smooth(method = "lm") +
  stat_cor(aes(label = after_stat(rr.label))) +
  labs(y = "Asthma Prevalence (%)",
       title = "AQI vs Asthma Prevalence by California County in 2020")
```

plot_2020

```
## `geom_smooth()` using formula = 'y ~ x'
```

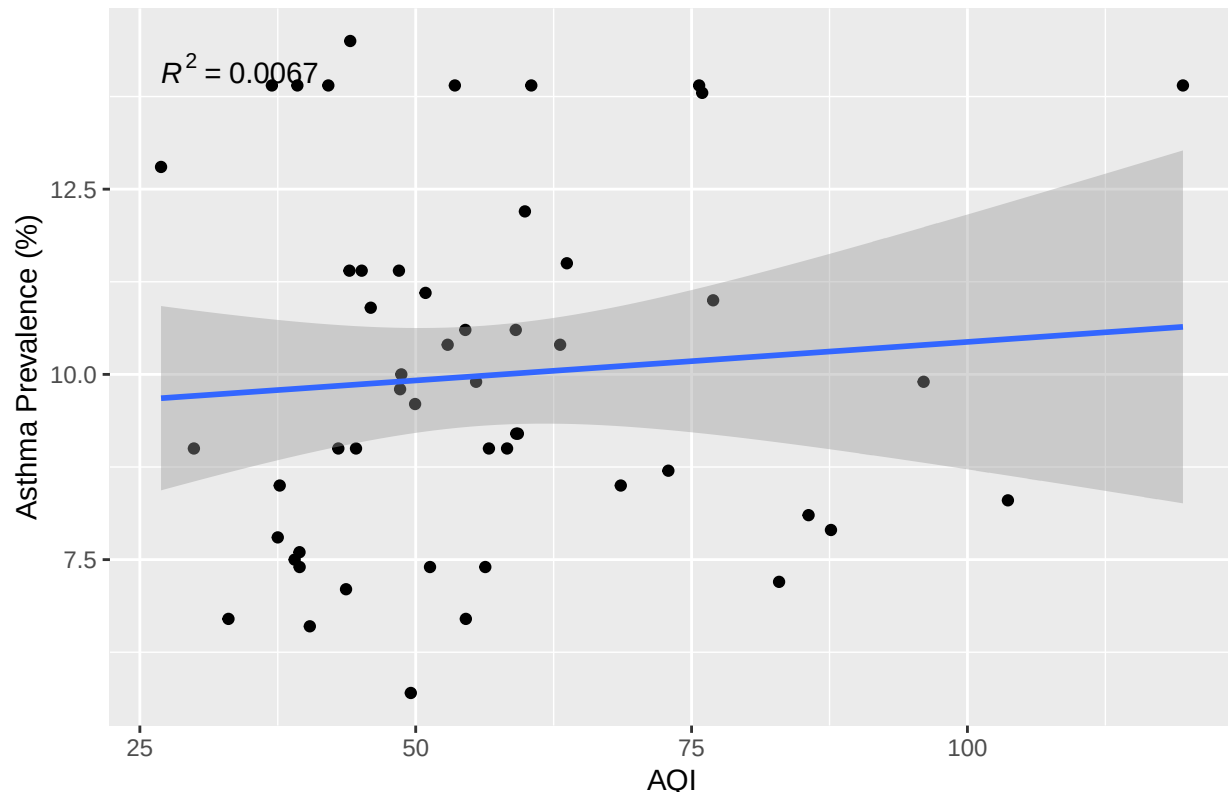


```
plot_2022 <- ggplot(data_2022, aes(AQI, `CURRENT PREVALENCE`)) +
  geom_point() +
  geom_smooth(method = "lm") +
  stat_cor(aes(label = after_stat(rr.label))) +
  labs(y = "Asthma Prevalence (%)",
       title = "AQI vs Asthma Prevalence by California County in 2022")
```

```
plot_2022
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

AQI vs Asthma Prevalence by California County in 2022

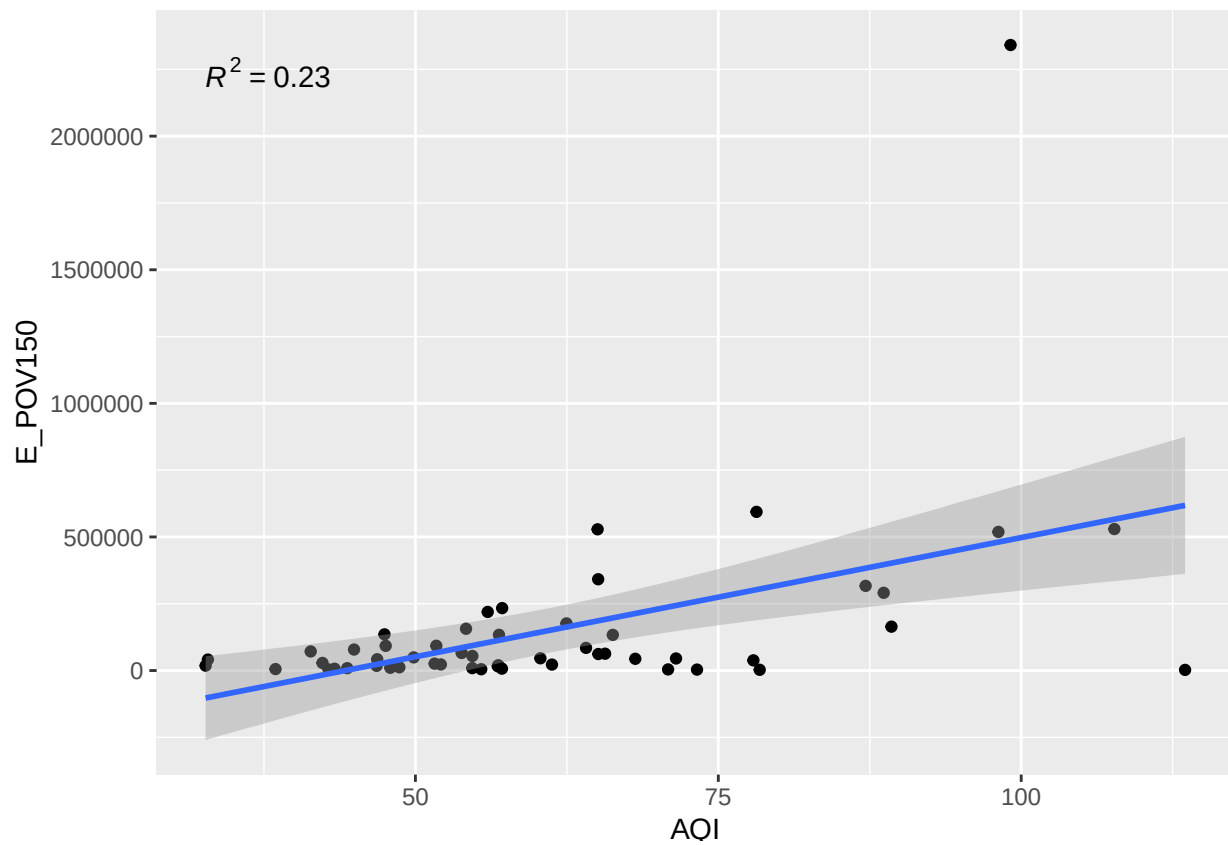


Plotting AQI and SVI metrics

```
plot_2020_SVI <- ggplot(data_2020_SVI, aes(AQI, E_P0V150)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  stat_cor(aes(label = after_stat(rr.label)))
```

```
plot_2020_SVI
```

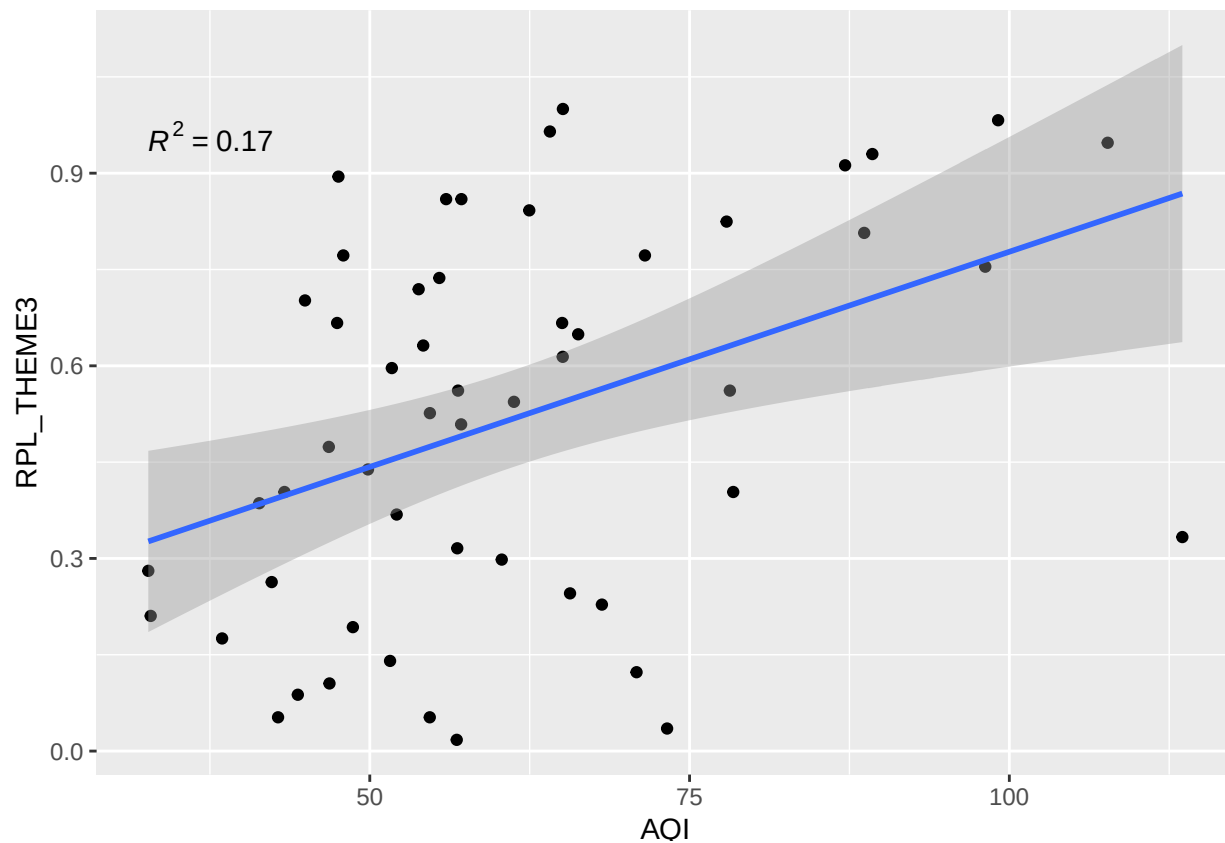
```
## `geom_smooth()` using formula = 'y ~ x'
```



```
plot_2020_RPL <- ggplot(data_2020_SVI, aes(AQI, RPL_THEME3)) +
  geom_point() +
  geom_smooth(method = "lm") +
  stat_cor(aes(label = after_stat(rr.label)))
```

```
plot_2020_RPL
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



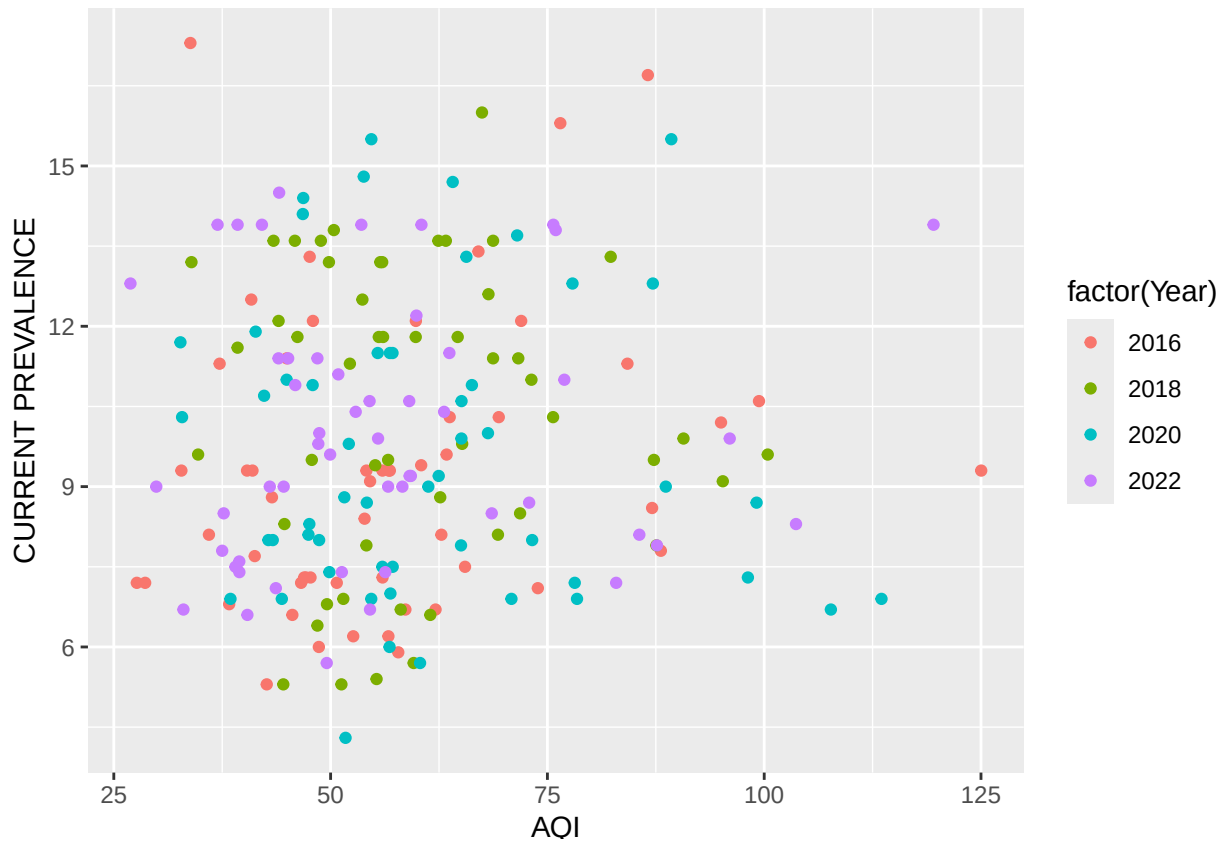
Combine data together

```
combined_data <- bind_rows(
  data_2016_SVI %>% mutate(Year = 2016),
  data_2018_SVI %>% mutate(Year = 2018),
  data_2020_SVI %>% mutate(Year = 2020),
  data_2022_SVI %>% mutate(Year = 2022)
)
```

Plot of current prevalence and aqi across years

```
combined_plot <- ggplot(combined_data, aes(AQI, `CURRENT PREVALENCE`,
  color = factor(Year))) + geom_point()
combined_plot
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
fires <- st_read("~/Downloads/California_Historic_Fire_Perimeters_-6273763535668926275")
```

DON'T RUN THIS CODE (will not work lol)

```
## Reading layer `California_Fire_Perimeters_(all)' from data source
## `~/Users/flannerymcnair/Downloads/California_Historic_Fire_Perimeters_-6273763535668926275'
## using driver `ESRI Shapefile'

## Warning in CPL_read_ogr(dsn, layer, query, as.character(options), quiet, : GDAL
## Message 1:
## /Users/flannerymcnair/Downloads/California_Historic_Fire_Perimeters_-6273763535668926275/California_
## contains polygon(s) with rings with invalid winding order. Autocorrecting them,
## but that shapefile should be corrected using ogr2ogr for example.

## Simple feature collection with 22810 features and 18 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: -13848330 ymin: 3833204 xmax: -12705610 ymax: 5255380
## Projected CRS: WGS 84 / Pseudo-Mercator
```

```
head(fires)
```

```
## Simple feature collection with 6 features and 18 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: -13213880 ymin: 4032811 xmax: -13137150 ymax: 4107141
## Projected CRS: WGS 84 / Pseudo-Mercator
##   YEAR_ STATE AGENCY UNIT_ID FIRE_NAME INC_NUM ALARM_DATE CONT_DATE CAUSE
```

```
## 1 2025 CA CDF LDF PALISADES 00000738 2025-01-07 2025-01-31 14
## 2 2025 CA CDF LAC EATON 00009087 2025-01-08 2025-01-31 14
## 3 2025 CA CDF ANF HUGHES 00250270 2025-01-22 2025-01-28 14
## 4 2025 CA CCO VNC KENNETH 00003155 2025-01-09 2025-02-04 14
## 5 2025 CA CDF LDF HURST 00003294 2025-01-07 2025-01-09 14
## 6 2025 CA CDF LAC LIDIA 00010648 2025-01-08 2025-01-11 14
## C_METHOD OBJECTIVE GIS_ACRES COMMENTS COMPLEX_NA
## 1 7 1 23448.900 <NA> <NA>
## 2 7 1 14056.300 <NA> <NA>
## 3 7 1 10396.800 <NA> <NA>
## 4 2 1 998.738 from OES Intel 24 <NA>
## 5 7 1 831.385 <NA> <NA>
## 6 7 1 347.704 <NA> <NA>
## IRWINID FIRE_NUM COMPLEX_ID DECADES
## 1 {A7EA5D21-F882-44B8-BF64-44AB11059DC1} <NA> <NA> 2020-January 2025
## 2 {72660ADC-B5EF-4D96-A33F-B4EA3740A4E3} <NA> <NA> 2020-January 2025
## 3 {994072D2-E154-434A-BB95-6F6C94C40829} <NA> <NA> 2020-January 2025
## 4 {842FB37B-7AC8-4700-BB9C-028BF753D149} <NA> <NA> 2020-January 2025
## 5 {F4E810AD-CDF3-4ED4-B63F-03D43785BA7B} <NA> <NA> 2020-January 2025
## 6 {E451AEAE-4177-4B78-BDE8-1E27B9D5D56C} <NA> <NA> 2020-January 2025
## geometry
## 1 MULTIPOLYGON (((-13193558 4...
## 2 MULTIPOLYGON (((-13146937 4...
## 3 MULTIPOLYGON (((-13197885 4...
## 4 MULTIPOLYGON (((-13211055 4...
## 5 MULTIPOLYGON (((-13187992 4...
## 6 MULTIPOLYGON (((-13164156 4...
```

```
fires$ALARM_DATE <- as.Date(fires$ALARM_DATE)
```

```
fires_clean <- fires %>%
  subset(select= -c(STATE, AGENCY, INC_NUM, C_METHOD, OBJECTIVE, COMMENTS,
    COMPLEX_NA, IRWINID, FIRE_NUM, COMPLEX_ID)) %>%
  filter(YEAR_ == "2022" | YEAR_ == "2020" | YEAR_ == "2018" | YEAR_ == "2016")
```

```
cali_counties <- st_read("~/Downloads/ca_counties/CA_Counties.shp")
```

```
## Reading layer `CA_Counties' from data source
## `~/Users/flannerymcnair/Downloads/ca_counties/CA_Counties.shp'
## using driver `ESRI Shapefile'
## Simple feature collection with 58 features and 19 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: -13857270 ymin: 3832931 xmax: -12705030 ymax: 5162404
## Projected CRS: WGS 84 / Pseudo-Mercator
```

```
occurrences_with_county <- st_join(
  fires_clean,
  cali_counties,
  join = st_intersects
)
```

```
write_csv(occurrences_with_county, here("Fires_per_county.csv"))
```

Load fire data

```

county_fires <- read_csv(here("Fires_per_county.csv"))

## Rows: 1682 Columns: 28
## -- Column specification -----
## Delimiter: ","
## chr   (14): UNIT_ID, FIRE_NAME, DECADES, STATEFP, COUNTYFP, COUNTYNS, GEOID, ...
## dbl   (12): YEAR_, CAUSE, GIS_ACRES, CSAFP, CBSAFP, METDIVFP, ALAND, AWATER, ...
## date   (2): ALARM_DATE, CONT_DATE
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

county_fires_clean <- county_fires %>%
  select(c(YEAR_, UNIT_ID, FIRE_NAME, ALARM_DATE, CONT_DATE, GIS_ACRES, DECADES,
           GEOID, NAME, ALAND, AWATER, Shape_Leng, Shape_Area, geometry))

```

Combining and cleaning fire data with AQI

```

fires_2022 <- county_fires %>%
  filter(YEAR_ == "2022")

fires_2022_counts <- as.data.frame(count(fires_2022, NAME)) %>%
  rename(COUNTY = NAME)

fires_aqi_2022 <- full_join(data_2022, fires_2022_counts, by = "COUNTY") %>%
  drop_na()

fires_2016 <- county_fires %>%
  filter(YEAR_ == "2016")

fires_2016_counts <- as.data.frame(count(fires_2016, NAME)) %>%
  rename(COUNTY = NAME)

fires_aqi_2016 <- full_join(data_2016, fires_2016_counts, by = "COUNTY")

```

Plot fires vs AQI and asthma

```

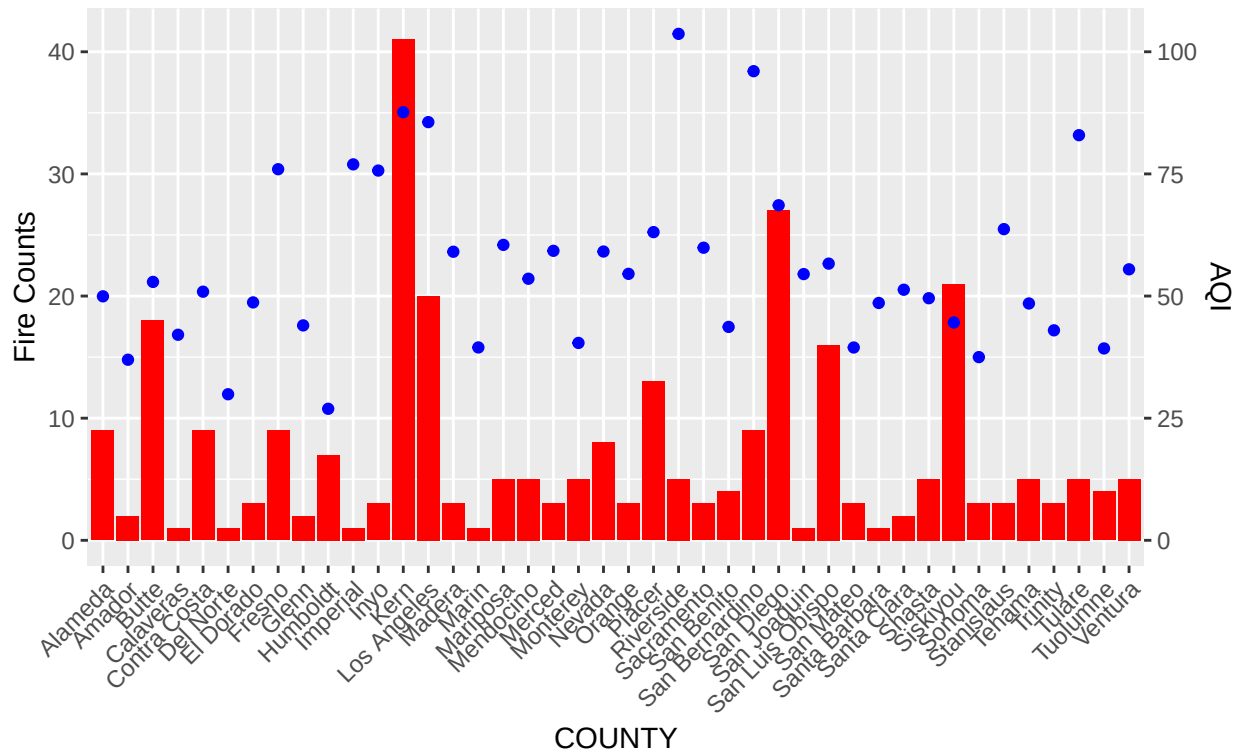
fire_plot <- ggplot(fires_aqi_2022, aes(x=COUNTY, y=n)) +
  geom_bar(stat = "identity", fill = "red") +
  geom_point(data = fires_aqi_2022, aes(x = COUNTY, y=AQI*0.4),
            color = "blue") +
  scale_y_continuous(name = "Fire Counts",
                    sec.axis = sec_axis(~ . / 0.4, name = "AQI")) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 1)) +
  labs(title = "Fire Counts and AQI per county in 2022",
       subtitle = "California")

fire_plot

```

Fire Counts and AQI per county in 2022

California



```
fire_plot_asthma <- ggplot(fires_aqi_2022, aes(x=COUNTY, y=n)) +
  geom_bar(stat = "identity", fill = "red") +
  geom_point(data = fires_aqi_2022, aes(x = COUNTY, y=`CURRENT PREVALENCE`),
    color = "blue") +
  scale_y_continuous(name = "Fires Counts",
    sec.axis = sec_axis(~ . / 0.4, name = "Asthma Prevalence")) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 1)) +
  labs(title = "Fire Counts and Asthma per county in 2022",
    subtitle = "California")
```

fire_plot_asthma

